TriFallNet: An AI System for Detecting Falls Early and Severity Assessment

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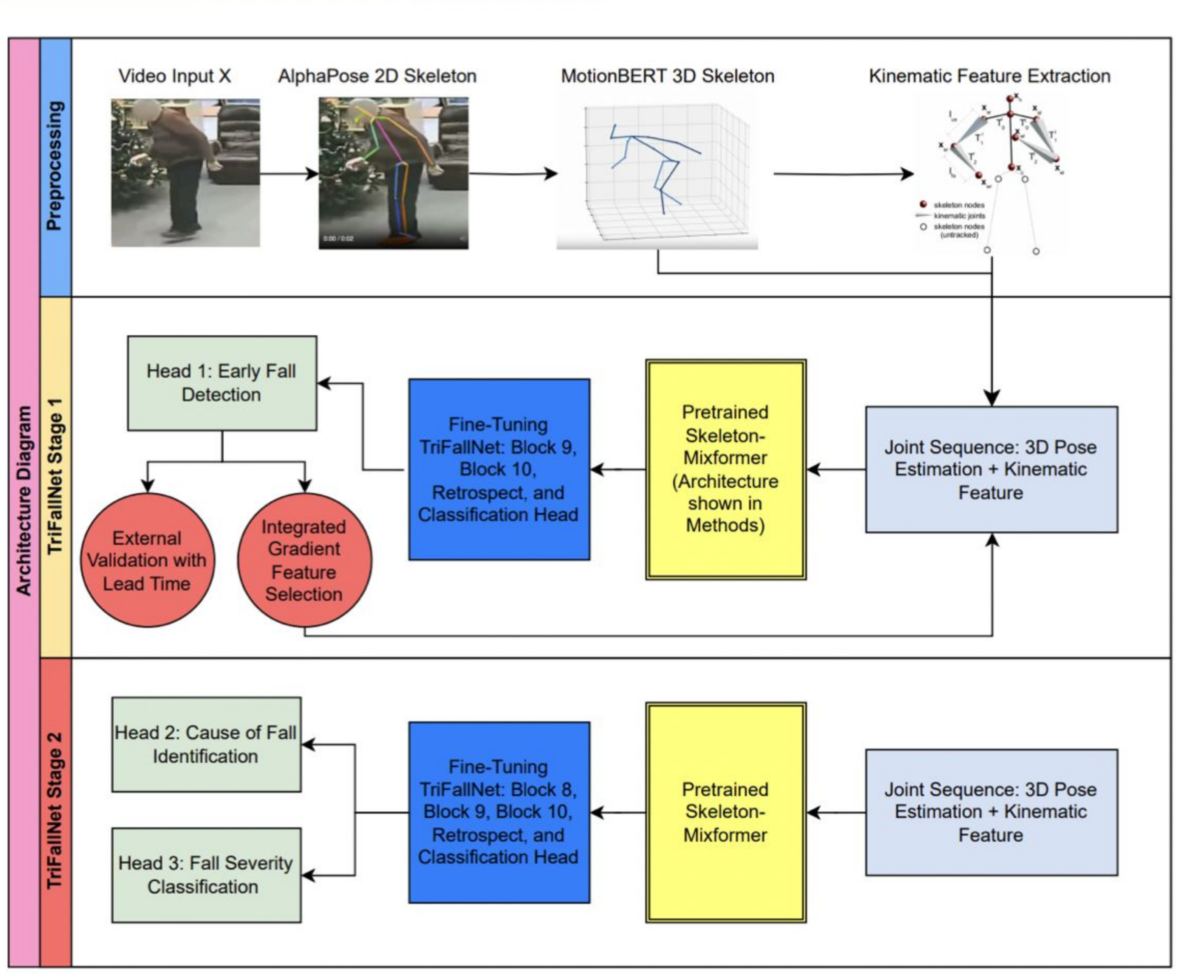
Abstract

- **Motivation**: Falls are a major health risk for older adults, but current systems rarely detect them early enough to prevent injury.
- **Approach**: **TriFallNet** uses 3D joint motion (MotionBERT) and kinematic features like center-of-mass velocity and torso inclination. A transformer architecture captures movement patterns to anticipate falls.
- **Key Results**: The model achieved an **AUC of 0.850**, detecting **85% of falls before impact** with an average lead time of **0.60 seconds**. This outperformed a baseline SVM, which was accurate frame-by-frame but weak at predicting falls in advance.
- **Future Work:** Generalize TriFallNet to alternative actions beyond falling to create a comprehensive framework to analyze **joint importance** in human mechanics**.**
- **Impact**: TriFallNet has applications in **elder care and clinical monitoring** for real-time fall prevention. Also, joint importance info assist future advancement in studies involving investigation of human actions.

Introduction

1. **Early Fall Detection**
2. **Joint Analysis**
3. **Cause Classification**
4. **Injury Severity Assessment**



Methodologies

Preprocessing: 3D Pose Estimation

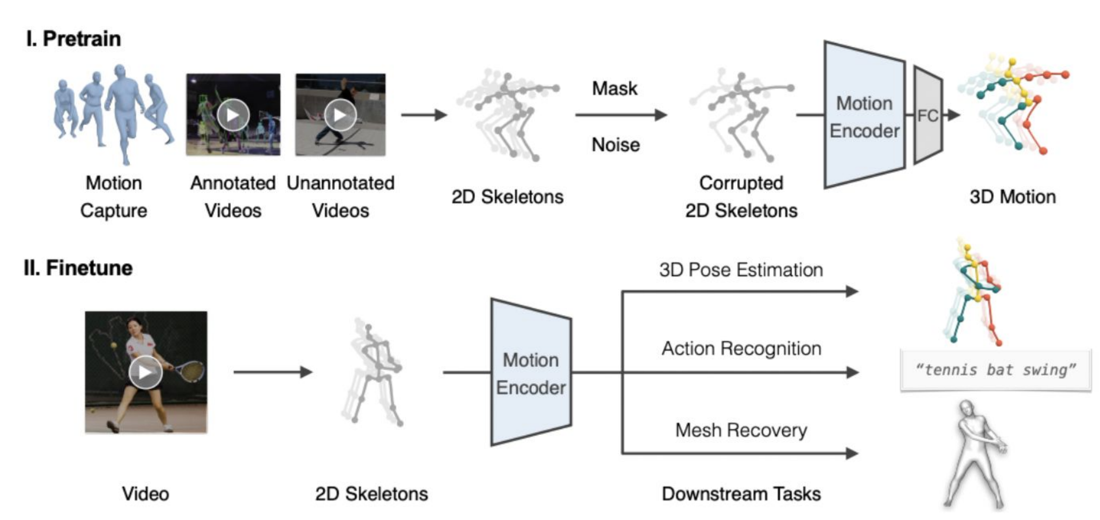


Figure 3: Framework of MotionBERT [9]

TriFallNet: Skeleton-Mixformer

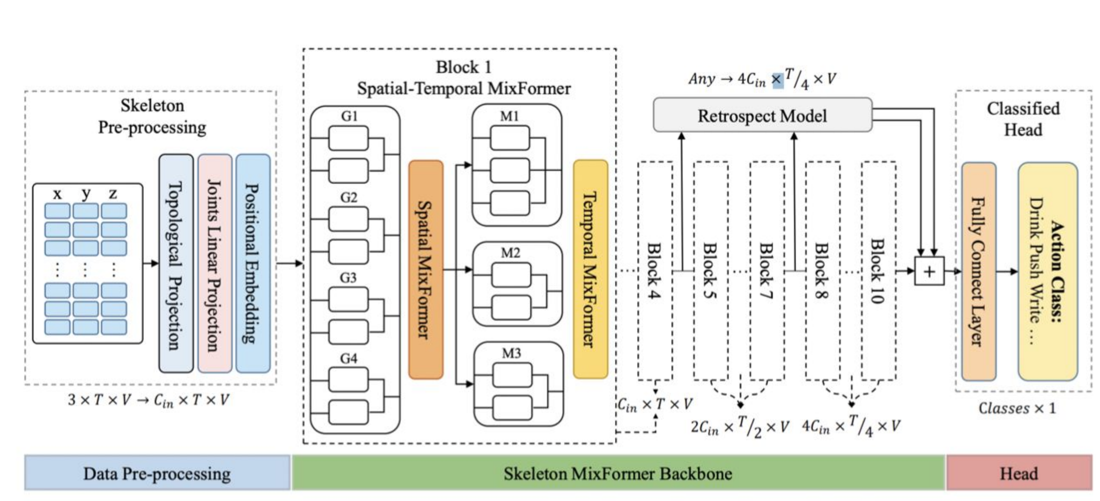


Figure 5: Overall Framework for Skeleton-MixFormer

Fall Detection Benchmark

- **Goal:** make cross-paper comparisons fair and measure early (pre-impact) warning quality.

- All clips are resampled to 25 Hz and resized to 256×256 RGB with aspect-ratio preserving letterboxing.
- Datasets are mapped to a common binary taxonomy (Fall vs ADL), with impact-aligned event windows retained for pre-impact evaluation.
- **Data split:** Le2i, URFD, and GMDCSA-24 are pooled for training/validation using subject-disjoint splits within each dataset.
- A single decision threshold is selected on the validation split and applied unchanged to all test evaluations.
- **Reporting:** results include clip-level metrics (F1, AUROC) and event-level metrics (event F1 under a fixed false-alarm budget, plus lead-time summary statistics).

Dataset	Modality	Views
Le2i (Charfi et al. 2013b)	RGB	Multi-room
URFD (Kwolek and Kepski 2014)	RGB	Single
GMDCSA-24 (Alam et al. 2024b)	RGB	Single/Multi
UP-Fall (Martínez-Villasenõr, Ponce, and et al. 2019)	RGB+wearables	Dual RGB
MultiCamFall (Auvinet et al. 2010b)	RGB	8 fixed

TriFallNet Result

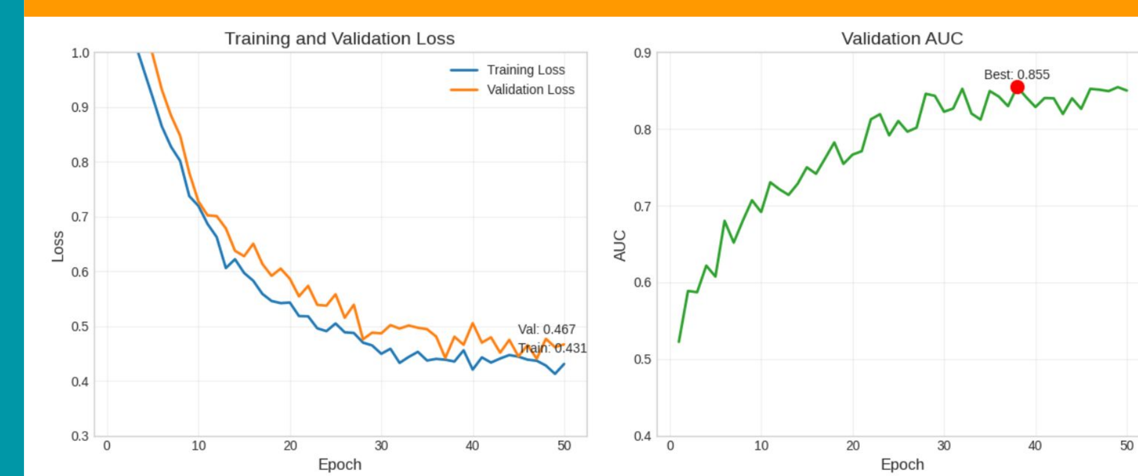


Figure 7: Training and validation loss and AUC curves for TriFallNet Stage 1 over 50 epochs. The best validation AUC attained is 0.855 during epoch 39 and the final validation loss is 0.467. Both curves converge smoothly and have a relatively small gap, indicating effective learning with minimal overfitting.

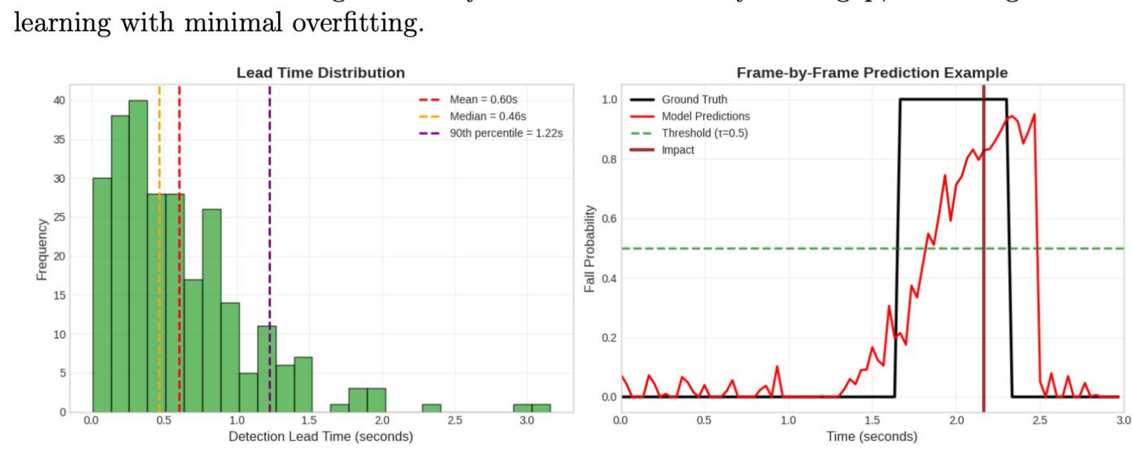
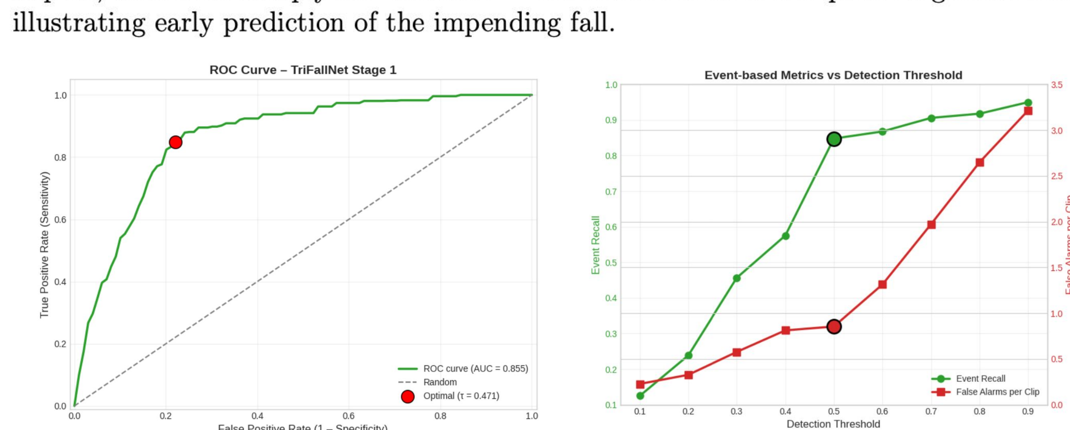


Figure 10: Temporal analysis of fall detection for successful events at $\tau = 0.5$. The left figure is a histogram of detection lead times across all clips in which the model signaled a fall before impact. The median lead time is 0.53 seconds and the 90th percentile is 1.18 seconds, demonstrating anticipatory detection. The right figure is an example of a per-frame fall-probability curve for one representative clip. The model's output hovers near 0.1 before impact, then rises sharply and crosses $\tau = 0.5$ around 0.4 seconds prior to ground contact, illustrating early prediction of the impending fall.



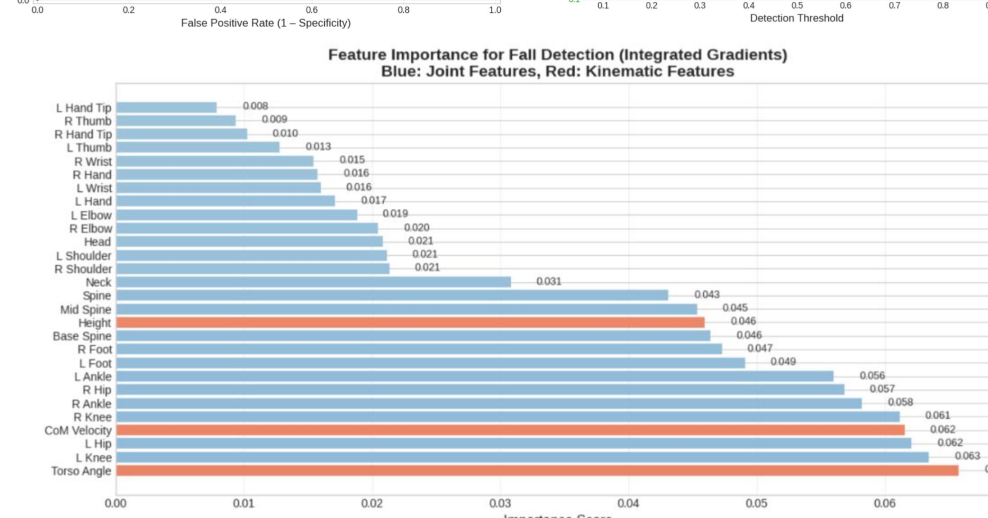


Figure 11: Integrated Gradients attribution for TriFallNet Stage 1. Reveals that lower body joints contributes more to fall detection compared to upper body joints. Also, kinematic features like CoM velocity and torso inclination angle serve as essential feature in TriFallNet.

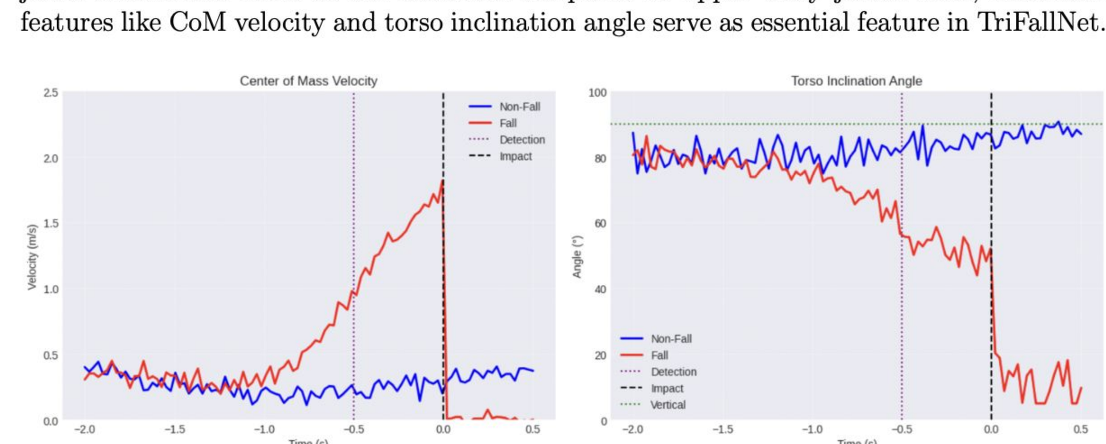


Figure 12: Visualization of two key kinematic features for fall detection: Center of Mass velocity and torso inclination angle. Fall instances show a steep rise in CoM velocity prior to ground impact followed by an abrupt drop post-impact. In contrast, non-fall instances maintain a relatively stable CoM velocity throughout.

Discussion

- **Performance**: TriFallNet achieved an AUC of 0.855, recalled 85% of falls, and predicted events **0.6s before impact**, compared to the SVM's 65% recall and no lead time.
- **Advances**: By combining **3D joint motion** with **kinematic features** (center-of-mass velocity, torso inclination), TriFallNet captured critical fall dynamics. Lower body joints and torso angle proved especially important for early detection.
- **Limitations**: Precision remained moderate, showing a need to reduce false alarms. Current datasets are small, often simulated, and lack age diversity, which limits generalizability.
- **Practicality**: The system runs at ~20 FPS (~50 ms latency), supporting real-time deployment. Balancing false alarms against missed falls remains key before clinical use.
- **Benchmark:** Our unified benchmark standardizes preprocessing and label mapping across datasets and reports fixed-threshold, event-centric early-warning metrics (including lead time) for fair comparison.

Conclusion

Future Work

- **Improve prediction:** Fusion of 3D pose and kinematic features delivers accurate fall prediction and early warning
  - Mean lead time >0.5 second enable rapid alerts and protective measure
- **Fall Mechanics:** New insights into which joints are important in falling to guide prevention strategies
- We provide a unified fall-detection benchmark with a single operating point and event-level early-detection metrics to support reproducible evaluation across datasets.